



INSTITUTO POLITÉCNICO NACIONAL
ESCUELA SUPERIOR DE CÓMPUTO



Introduction to Cryptography

Lab 09: Multiplication on $GF(2^n)$

May 7, 2026

1. Pseudocode for multiplication on $GF(2^n)$

Do the following exercises **by your own. Write by hand** in your notebook or tablet the pseudocode of the algorithms, using the mathematical notation given in the class. Upload to Microsoft Teams, before the session ends, screenshots of your pseudocode and some examples to see that your algorithm works properly.

1. Design an algorithm to multiply two elements $a, b \in GF(2^n)$. Your algorithm must receive as input the binary representation of a and b . The irreducible polynomial $m(x)$ for $GF(2^n)$ will be fixed, also it will be represented as a binary string. Your algorithm will return $a * b \bmod m(x)$ in binary. You can only use bitwise operations.
2. Do at least 5 examples for different binary finite fields, to see that your algorithm works properly.

2. Programming exercises

Each student must do the following exercises using C/C++ programming language and bitwise operators.

1. Design and implement a function that receives as parameter an element $f(x) \in GF(2^{16})$. Use an unsigned short variable to store the coefficients of $f(x)$ i.e. use the binary representation of $f(x)$. Your function must compute $x * f(x) \bmod (x^{16} + x^5 + x^3 + x + 1)$ and print the value of $f(x)$ and the result $x * f(x)$ in hexadecimal.
2. Design and implement a function that receives as parameter two elements $f(x), g(x) \in GF(2^{16})$. Use an unsigned char variables to store the coefficients of $f(x)$ and $g(x)$ i.e. use the binary representation of $f(x)$ and $g(x)$. Your function must compute $f(x) * g(x) \bmod (x^{16} + x^5 + x^3 + x + 1)$ and print the values of $f(x), g(x)$ and the result of multiplication $f(x) * g(x) \bmod (x^{16} + x^5 + x^3 + x + 1)$ in hexadecimal.
3. Test both functions with at least 5 different values of $f(x)$ and $g(x)$

3. Products

Every student must write his/her own document to present the results. Your document must include

1. Personal information, date of the lab session and the topic that we are studying in this lab session.
2. Source code you implemented for exercises 1, 2 and 3 of Section 2.
3. Screenshots of your programs running.